# Chapter 3 Classification of Elements and Periodicity in Properties

## Modern Periodic Law, Nomenclature of Elements

**1.** The IUPAC name of an element with atomic number 119 is. *(2022)*

a. Ununcotium b. Ununennium
c. Unnilennium d. Unununnium

**2.** Identify the incorrect match *(2020)*

| | Name | | IUPAC Official Name |
|---|---|---|---|
| (A) | Unnilunium | (i) | Mendelevium |
| (B) | Unniltrium | (ii) | Lawrencium |
| (C) | Unnilhexium | (iii) | Seaborgium |
| (D) | Unununnium | (iv) | Darmstadtium |

a. (B), (ii) b. (C), (iii) c. (D), (iv) d. (A), (i)

## Electronic Configurations and Types of Elements (s, p, d and f-Blocks)

**3.** The element Z = 114 has been discovered recently. It will belong to which of the following family group and electronic configuration? *(2017-Delhi)*

a. Nitrogen family, [Rn] $5f^{14}6d^{10}7s^{2}7p^{6}$
b. Halogen family, [Rn] $5f^{14}6d^{10}7s^{2}7p^{5}$
c. Carbon family, [Rn] $5f^{14}6d^{10}7s^{2}7p^{2}$
d. Oxygen family, [Rn] $5f^{14}6d^{10}7s^{2}7p^{4}$

**4.** The number of d-electrons in $Fe^{2+}$ (Z = 26) is not equal to the number of electrons in which one of the following? *(2015)*

a. p-electrons in Cl (Z = 17)
b. d-electrons in Fe (Z = 26)
c. p-electrons in Ne (Z = 10)
d. s-electrons in Mg (Z = 12)

## Periodic Trends in Properties of Elements

**5.** The correct decreasing order of atomic radii (pm) of Li, Be, B and C and C is *(2024 Re)*

a. Be > Li > B > C b. Li > Be > B > C
c. C > B > Be > Li d. Li > C > Be > B

**6.** Arrange the following elements in increasing order of electronegativity: *(2024)*
N, O, F, C, Si
Choose the correct answer from the options given below:
a. O < F < N < C < Si b. F < O < N < C < Si
c. Si < C < N < O < F d. Si < C < O < N < F

**7.** Arrange the following elements in increasing order of first ionization enthalpy: *(2024)*
Li, Be, B, C, N
Choose the correct answer from the options given below:
a. Li < Be < C < B < N b. Li < Be < N < B < C
c. Li < Be < B < C < N d. Li < B < Be < C < N

**8.** The element expected to form largest ion to achieve the nearest noble gas configuration is: *(2023)*

a. Na b. O c. F d. N

**9.** Given below are two statements: one is labelled as *(2023-Manipur)*
**Assertion (A)** and the other is labelled as **Reason (R)**
**Assertion (A):** Ionisation enthalpy increases along each series of the transition elements from left to right. However, small variations occur.
**Reason (R):** There is corresponding increase in nuclear charge which accompanies the filling of electrons in the inner d-orbitals.
In the light of the above statements, choose the **most appropriate** answer form the options given be:
a. **(A)** is correct but **(R)** is not correct.
b. **(A)** is not correct but **(R)** is correct.
c. Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**.
d. Both **(A)** and (**R**) are correct **(R)** is **not** the correct explanation of **(A)**.

**10.** Which one of the following represents all isoelectronic species? *(2023-Manipur)*

a. $Na^{+}$, $Cl^{-}$, $O^{-}$, $NO^{+}$ b. $N_2O$, $N_2O_4$, $NO^{+}$, NO
c. $Na^{+}$, $Mg^{2+}$, $O^{-}$, $F^{-}$ d. $Ca^{2+}$, Ar, $K^{+}$, $Cl^{-}$

**11.** The **correct** sequence given below containing neutral, acidic, basic and amphoteric oxide each, respectively, is *(2023-Manipur)*

a. NO, ZnO, $CO_2$, CaO b. ZnO, NO, CaO, $CO_2$
c. NO, $CO_2$, ZnO, CaO d. NO, $CO_2$, CaO, ZnO

**12.** The correct order of first ionization enthalpy for the given four elements is: *(2022 Re)*

a. C < F < N < O b. C < N < F < O
c. C < N < O < F d. C < O < N < F

**13.** If first ionization enthalpies of elements X and Y are 419 kJ $mol^{-1}$ and 590 kJ $mol^{-1}$, respectively and second ionization enthalpies of X and Y are 3069 kJ $mol^{-1}$ and 1145 kJ $mol^{-1}$, respectively. Then **correct statement** is: *(2022 Re)*

a. Both X and Y are alkaline earth metals.

b. X is an alkali metal and Y is an alkaline earth metal.

c. X is an alkaline earth metal and Y is an alkali metal.

d. Both X and Y are alkali metals.

**14.** From the following pairs of ions, which one is not an iso-electronic pair? *(2021)*

a. $Na^+$, $Mg^{2+}$ b. $Mn^{2+}$, $Fe^{3+}$

c. $Fe^{2+}$, $Mn^{2+}$ d. $O^{2-}$, $F^-$

**15.** For the second period elements, the correct increasing order of first ionisation enthalpy is: *(2019)*

a. Li < Be < B < C < N < C < F < Ne

b. Li < B < Be < C < O < N < F < Ne

c. Li < B < Be < C < N < O < F < Ne

d. Li < Be < B < C < O < N < F < Ne

**16.** In which of the following options, the order of arrangement does not agree with the variation of property indicated against it? *(2016 - I)*

a. Li < Na < K < Rb (increasing metallic radius)

b. $Al^{3+} < Mg^{2+} < Na^+ < F^-$ (increasing ionic size)

c. B < C < N < O (increasing first ionization enthalpy)

d. I < Br < Cl < F (increasing electron gain enthalpy)

**17.** The formation of the oxide ion, $O^{2-}$ (g) from oxygen atom requires first an exothermic step and then an endothermic step as shown below:

$O(g) + e^- \rightarrow O^-(g)$ ; $\Delta_f H^o = -141$ kJ $mol^{-1}$

$O^-(g) + e^- \rightarrow O^{2-}(g)$ ; $\Delta_f H^o = +780$ kJ $mol^{-1}$

Thus, process of formation of $O^{2-}$ in gas phase is unfavourable even though $O^{2-}$ is isoelectronic with neon. It is due to the fact that, *(2015 Re)*

a. $O^-$ ion has comparatively smaller size than oxygen atom

b. Oxygen is more electronegative

c. Addition of electron in oxygen results in larger size of the ion

d. Electron repulsion outweighs the stability gained by achieving noble gas configuration

**18.** The species Ar, $K^+$ and $Ca^{2+}$ contain the same number of electrons. In which order do their radii increase? *(2015)*

a. $Ca^{2+} < Ar < K^+$ b. $Ca^{2+} < K^+ < Ar$

c. $K^+ < Ar < Ca^{2+}$ d. $Ar < K^+ < Ca^{2+}$

**19.** $Be^{2+}$ is isoelectronic with which of the following ions? *(2014)*

a. $Li^+$ b. $Na^+$

c. $Mg^{2+}$ d. $H^+$

**20.** Which of the following orders of ionic radii is correctly represented? *(2014)*

a. $Na^+ > F^- > O^{2-}$ b. $O^{2-} > F^- > Na^+$

c. $Al^{3+} > Mg^{2+} > N^{3-}$ d. $H^- > H^+ > H$

## Answer Key

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **1.** (b) | **2.** (c) | **3.** (c) | **4.** (a) | **5.** (b) | **6.** (c) | **7.** (d) | **8.** (d) | **9.** (c) | **10.** (d) |
| **11.** (d) | **12.** (d) | **13.** (b) | **14.** (c) | **15.** (b) | **16.** (c, d) | **17.** (d) | **18.** (b) | **19.** (a) | **20.** (b) |